✈ Airplane Management System

About the Project

The Airplane Management System is a web-based application developed using Python and Django. It helps manage flight information, passenger bookings, ticket prices, destinations, and ticket status through a centralized system.

The project includes an administrative interface for managing flights and bookings, user authentication, and an Airline Management Information System (MIS) dashboard.

Features

Flight Management: Add and manage flight numbers, airline names, airplane models, and routes.
Destination Management: Store departure cities and arrival destinations.
Ticket Management: Manage ticket prices, booking references, and ticket status.
Passenger Management: Associate bookings with registered user accounts.
Booking Management: Create and track passenger bookings.
Seat Availability: Track available seats for scheduled flights.
User Authentication: Support user login through Django authentication.
Admin Panel: Manage flights and bookings using Django Admin.
MIS Dashboard: Display flight statistics, booking information, seat availability, and recorded paid-booking revenue.

Technologies Used

Python
Django
HTML
SQLite
Django Authentication
Django Admin

Project Modules

- Flight Management
- Passenger and User Management
- Ticket Booking Management
- Ticket and Payment Status
- Airline MIS Dashboard
- Administrative Management

Installation and Setup

1. Clone the repository

```
git clone https://github.com/YOUR-USERNAME/airplane-management-system.git
cd airplane-management-system
```

2. Create a virtual environment

```
python -m venv .venv
```

3. Activate the environment

Windows CMD:

```
.venv\Scripts\activate
```

4. Install Django

```
python -m pip install django
```

5. Apply database migrations

```
python manage.py migrate
```

6. Create an administrator account

```
python manage.py createsuperuser
```

7. Run the development server

```
python manage.py runserver
```

Application URLs

Main Website: http://127.0.0.1:8000/
User Login: http://127.0.0.1:8000/accounts/login/
Admin Panel: http://127.0.0.1:8000/admin/
MIS Dashboard: http://127.0.0.1:8000/mis/

Future Enhancements

Passenger registration
Online payment gateway integration
Ticket cancellation and refund management
Electronic ticket generation
Flight search and filtering
Email booking confirmations
Improved dashboard reports and analytics
Automated seat allocation

Disclaimer

This project is intended for educational and demonstration purposes. It is not a production-ready airline reservation or payment system.

Author

Developed as a Python and Django web development project.